Head of the Lake 2026: Tate 4+ race analysis

Sunday 1 November 2026, Head course, crew off the start at about 1:06 PM. Source: GPS file 20261101_0106PM_HOTL_Tate_4+.gpx (one point every 2 seconds, no stroke rate). Wind was light, about 4 kt from 310°, so it barely shaped the race.

The race in one paragraph

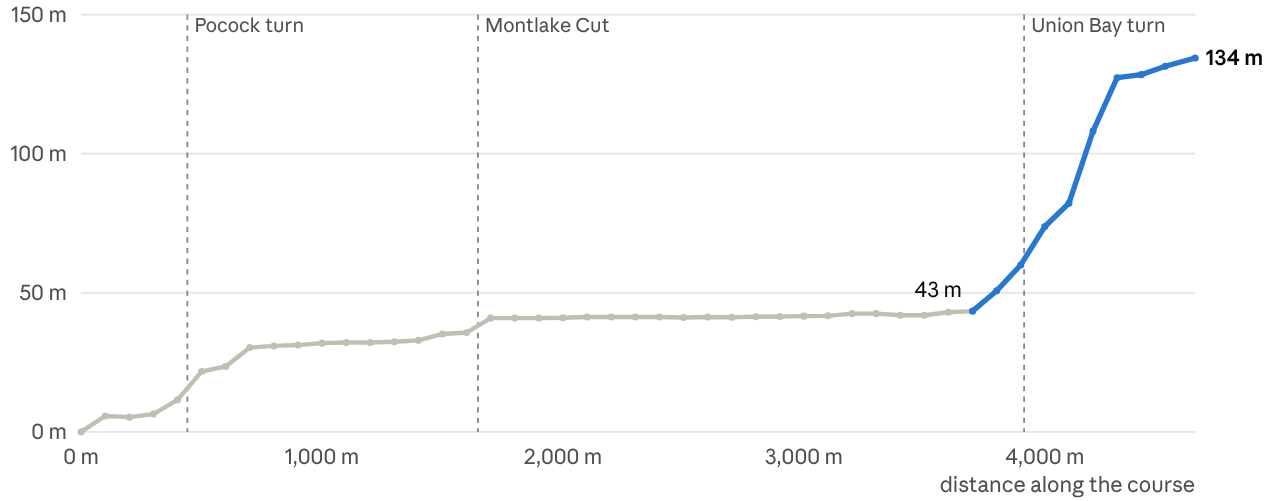
The crew covered the course in **19:09** (CoxPilot Replay says 19:07; the 2 s gap comes from where each one places the start and finish crossings). The boat rowed **4,759 m** against **4,624 m** on the shortest smooth line, so it rowed **about 134 m extra**. That is about **10 boat lengths**, roughly **17 strokes at the crew's 7.7 m per stroke**, or **about 32 seconds** at the crew's average boat speed (2:00.7 /500 m). Steering on the straights was very good. Most of the lost water was at the Union Bay turn and on the run home, with a smaller loss off the start.

Where the metres went

Leg	Time	Boat split	Rowed	Smooth line	Extra	Off line avg / max
Start to Pocock turn	1:50.1	2:00.1	458 m	441 m	+17 m	58 / 75 m (to port)
Pocock turn to Montlake Cut	4:51.7	1:58.8	1,227 m	1,206 m	+22 m	16 / 31 m
Montlake Cut to Union Bay turn	9:22.1	2:02.6	2,293 m	2,267 m	+25 m	10 / 28 m
Union Bay turn to finish	3:05.1	1:58.6	780 m	710 m	**+70 m**	40 / 63 m (to starboard)

Two-thirds of the extra water came at the Union Bay turn

Extra metres rowed beyond the shortest smooth line, added up along the course.
134 m in all (about 10 boat lengths); 91 m of it from the Union Bay turn home.



GPS file 20261101_0106PM_HOTL_Tate_4+.gpx · every 100 m of the shortest smooth line

The flat stretch from the Montlake Cut to Union Bay is near-perfect steering. The steps are the start and Pocock turn (about 30 m) and the Union Bay turn and run home (about 90 m).

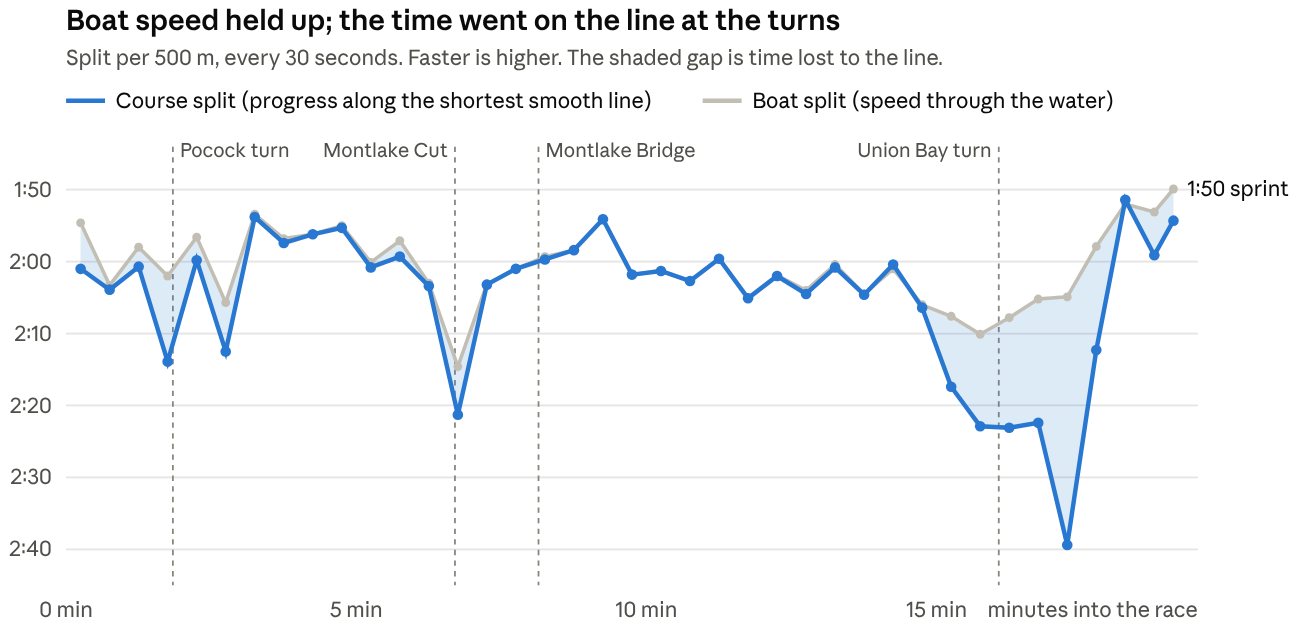
The Union Bay turn and run home alone cost about 70 to 90 m (5 to 7 lengths, around 20 seconds). Everything from the Montlake Cut to the start of the Union Bay turn was within a few metres of ideal.

Splits by the 500

Course split is time over each 500 m of the shortest line, so it shows the cost of a wide line. Boat split is time over the water the boat actually rowed, so it shows boat speed. When the two are far apart, the line cost time, not the rowing.

Course metres	Course split	Boat split	Note
0 to 500	2:04.7	1:59.5	Wide off the start, around Pocock turn
500 to 1,000	2:00.2	1:57.8	Under University Bridge, into Portage Bay
1,000 to 1,500	1:57.8	1:57.0	Fastest steady piece
1,500 to 2,000	2:06.9	2:05.4	Speed dip into the Montlake Cut

Course metres	Course split	Boat split	Note
2,000 to 2,500	1:57.4	1:57.4	Out of the Cut, strong
2,500 to 3,000	2:02.9	2:02.8	
3,000 to 3,500	2:02.4	2:02.3	
3,500 to 4,000	2:15.7	2:07.5	Fade plus wide into Union Bay turn
4,000 to 4,500	2:12.3	1:58.7	Good push, but wide line home
4,500 to finish (124 m)	1:55.9	1:53.2	Sprint, last 30 s at about 1:50



GPS file 20261101_0106PM_HOTL_Tate_4+.gpx · 30 s blocks, start to finish

Where the blue line pulls away from the grey one, the crew was rowing fine but steering a longer line. The biggest gap is the 2 minutes through and out of the Union Bay turn.

Steering on the straights

There is no snaking. Once the slow trend of the line is taken out, side-to-side weave was under 1 m on every straight (0.7 m in the Montlake Cut, 0.6 m along Union Bay east), which is about the size of GPS noise. The file has a point every 2 seconds, so a wobble inside a

single stroke would not show, but there is no sign of the boat drifting off and being pulled back with big rudder.

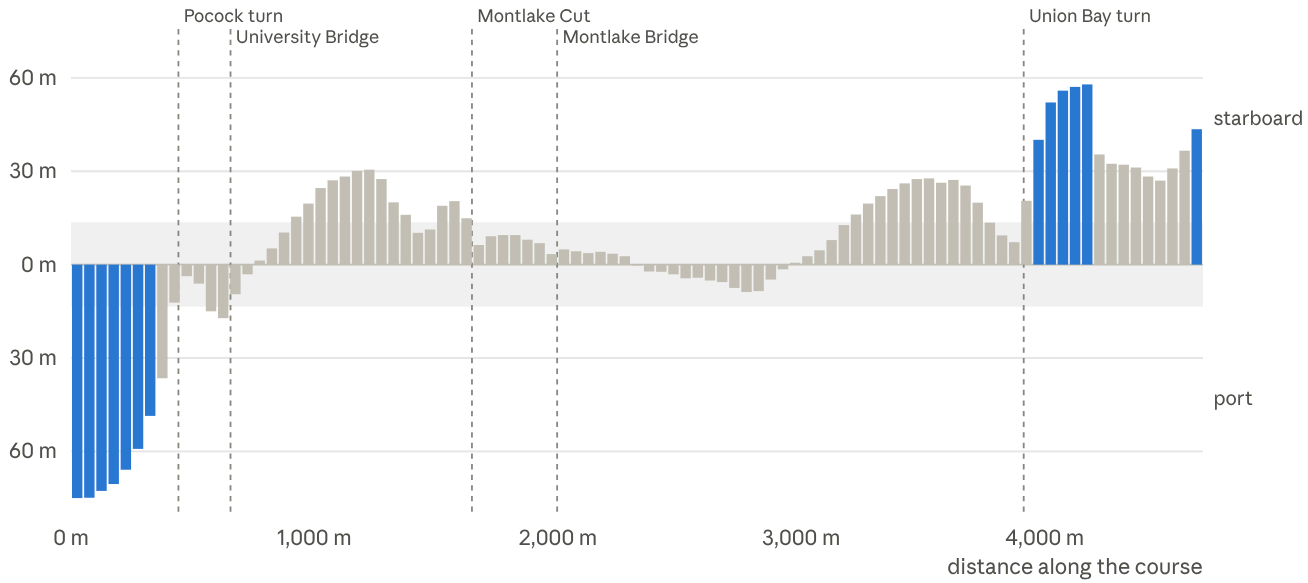
Headings held: about 36° up from the start, about 129° through Portage Bay, 91° through the Cut and along Union Bay, and about 280° on the run home.

Turns and bridges

- **Pocock turn:** taken about 38 m off the turn buoy, against about 30 m on the ideal line. Clean, a little wide, about 1 to 2 lengths.
- **University Bridge (2:44 in):** came through 81% of the way across the span, where the ideal line is at 87%. That is about 15 m off, close to ideal.
- **Montlake Cut entry:** the line was good (about 19 m off the corner), but boat speed dropped. From about 6:00 to 7:00 in, the boat split went from about 1:57 to 2:14, which cost about 6 to 7 seconds. The line does not explain it, so it was something in the boat or traffic (see question below).
- **Montlake Bridge (8:09 in):** dead centre, 52% across the span against 49% ideal.
- **Union Bay turn (about 16:00 in):** closest to the turn buoy was about 25 m, against about 11 m on the ideal line. The bigger cost came on the way out: the boat swung up to about 55 m outside the line, then ran home 30 to 45 m north of the buoy line and crossed the middle of the finish line instead of the south end. Boat speed there was good (1:52 to 1:58), so this was pure line.
- **Start:** the crew crossed the start line at its west end and ran up the west side, 60 to 75 m to port of the ideal line, before closing in at Pocock. Because the line was straight, it cost only about 17 m (just over a length).

The crew was widest off the start and coming out of the Union Bay turn

Distance from the shortest smooth line every 50 m. Widest: 75 m to port at 25 m, 58 m to starboard at 4,175 m.
Shaded band = within one boat length (13.5 m) of the line.



GPS file 20261101_0106PM_HOTL_Tate_4+.gpx · median every 50 m, track lined up on the traced map

Through the Montlake Cut and along Union Bay the boat stayed inside one length of the ideal line for over a kilometre. Being wide off the start cost little because the line was straight; being wide out of the Union Bay turn cost the most because the crew stayed out there all the way home.

Pacing

The crew settled well (1:57 to 1:59 boat split for the first 1,500 m) and came out of the Cut strong (1:54 at about 9:00). From about 14:30 to 17:00 in, before and through the Union Bay turn, boat split faded to 2:05 to 2:10. After the turn the crew pushed hard (1:52 to 1:58) and sprinted at about 1:50. The strongest rowing came after the most expensive steering.

Stroke rate and run per stroke

The rate never dropped. Every slow stretch came from a shorter run per stroke, not a lower rate. Source: the cox box file CoxBox_011010824_20261101_0106PM.csv (656 strokes, average rate 32½, 7.7 m per stroke).

Stretch	Time in	Rate (spm)	Run per stroke	Boat split
Start, Pocock turn, University Bridge	0:00 to 3:00	31.4	8.0 m	2:00.0
Portage Bay	3:00 to 6:00	31.8	8.1 m	1:56.8
Montlake Cut entry	6:00 to 8:00	32.3	7.5 m	2:05.5
Through the Cut and along Union Bay	8:00 to 14:30	32.4	7.7 m	2:01.6
Before and through Union Bay turn	14:30 to 17:00	33.3	7.2 m	2:06.8
Run home and sprint	17:00 to finish	34.4	7.5 m	1:57.2

At the Montlake Cut entry and again before the Union Bay turn the rate went up while the run per stroke fell by about half a metre to a metre. That is the boat being rushed or the legs going soft, not the crew easing off. The calls that fix it are about length and legs, not rate.

What to practise

1. **Union Bay turn, held tight on the buoys.** Call the turn early, hold about one length off the orange buoys on port all the way round, and hold the rudder until the bow points at the south end of the finish (about 280°). Worth about 5 lengths. The Turns drill with a long 110° port turn is the closest match.
2. **Start on the starboard (east) end of the start line** and point straight at the inside of Pocock turn. Worth about 1 length.
3. **Keep the length through the Montlake Cut.** The rate held at about 32 there, but the run per stroke fell from 8.1 m to 7.5 m. Plan a length call for the Cut entry ("long and strong, send it") rather than more rate.
4. **The 14 to 16 minute stretch.** The rate climbed to 33 here but the run per stroke dropped to 7.2 m, the shortest of the race. Plan a legs call before Union Bay turn ("push the feet, long through the turn") so the fade does not stack on top of the turn, and keep the rate where it is.

How sure are these numbers

- The course in CoxPilot is traced from the official map, which sits about 60 m off true GPS. Both CoxPilot and this analysis slide the track about 59 to 63 m to line it up; after that the track sits within about 5 m of the line in the Montlake Cut, where the channel is narrow, which is a good sign the line-up is right.
- Rowed distance comes straight from GPS and does not depend on the line-up. The smooth-line distance does depend on the traced map, which can be off by tens of metres in places. Treat leg extras as within about ±10 m.
- Stroke rate and stroke counts come from the cox box file (656 strokes, average rate 32½, 7.7 m per stroke). Its track matches the GPX file to within a second on time and a metre on distance.

Replay tab check

The file loads in CoxPilot's Replay tab with no errors. It picks the Head course, lines the track up by moving it 59 m, and shows 19:07, 1:59 average, +127 m extra. The legs table shows the same picture (Union Bay turn to finish +76 m). The cox box CSV imports the same way and fills the Rate column: 31, 32, 33 and 34 by leg.